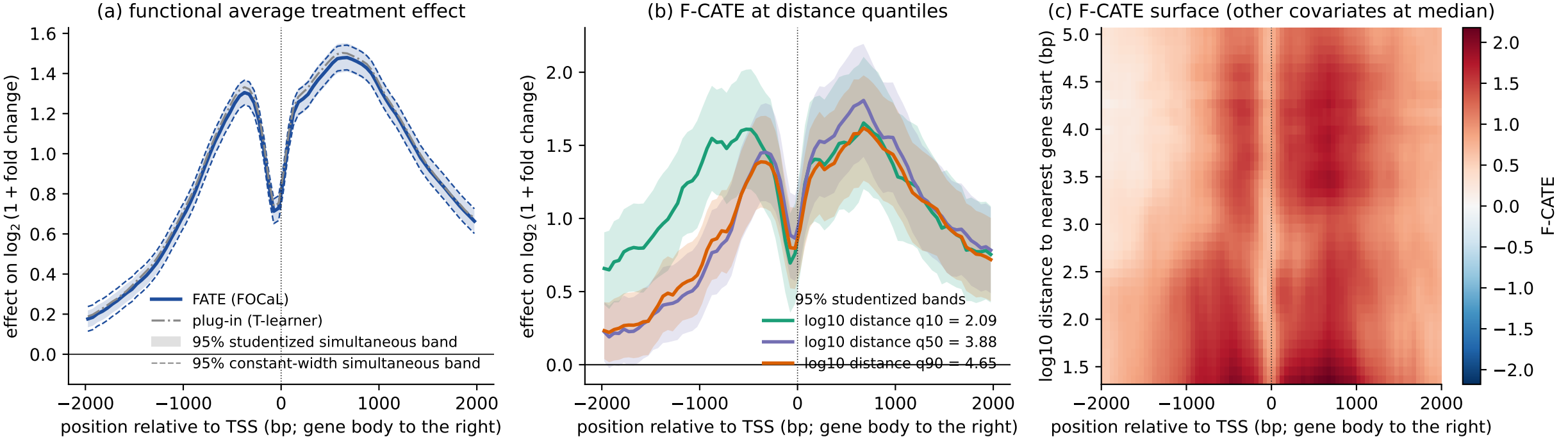

(a) functional average treatment effect
effect on $\log_2(1 + \text{fold change})$
position relative to TSS (bp; gene body to the right)
FATE (FOCaL)
plug-in (T-learner)
95% studentized simultaneous band
95% constant-width simultaneous band
(b) F-CATE at distance quantiles
effect on $\log_2(1 + \text{fold change})$
position relative to TSS (bp; gene body to the right)
95% studentized bands
log10 distance q10 = 2.09
log10 distance q50 = 3.88
log10 distance q90 = 4.65
(c) F-CATE surface (other covariates at median)
log10 distance to nearest gene start (bp)
position relative to TSS (bp; gene body to the right)
F-CATE